

วิธีการรันและผลลัพธ์

1.เปิด Docker containers ทั้งหมด

- docker compose up -d

```
boomn@DESKTOP-IU0AVN0 MINGW64 ~/OneDrive/Documents/GitHub/DADS6002/code (main)
$ docker compose up -d
[+] up 12/12
✓ Network code_default Created
✓ Container hive-metastore-postgresql Started
✓ Container historyserver Started
✓ Container resourcemanager Started
✓ Container namenode Started
✓ Container spark-master Started
✓ Container datanode Started
✓ Container nodemanager Started
✓ Container hive-metastore Started
✓ Container spark-worker Started
✓ Container hive-server Started
✓ Container hue Started

what's next:
  Filter, search, and stream logs from all your Compose services
  in one place with Docker Desktop's Logs view. docker-desktop://dashboard/logs?appId=code

boomn@DESKTOP-IU0AVN0 MINGW64 ~/OneDrive/Documents/GitHub/DADS6002/code (main)
$
```

2.รัน MapReduce Job

- MSYS_NO_PATHCONV=1 ./execute.sh

```

2026-06-02 15:35:04 INFO hadoop.MING64@-jondr-live/documents/gi/pub/DAOS6002/code (main)
  mvsv_00_PATHCONV-1 -/execute.sh
Successfully copied 319k (transferred 2.56kb) to namenode:/mapper.py
Successfully copied 986k (transferred 2.56kb) to namenode:/reducer.py
Successfully copied 717k (transferred 2.56kb) to namenode:/input.txt

What's next:
  Try Docker Debug for seamless, persistent debugging tools in any container or image - docker debug namenode
  Learn more at https://docs.docker.com/go/debug-cli/
  mkdir: cannot create directory '/job_files': File exists

What's next:
  Try Docker Debug for seamless, persistent debugging tools in any container or image - docker debug namenode
  Learn more at https://docs.docker.com/go/debug-cli/

What's next:
  Try Docker Debug for seamless, persistent debugging tools in any container or image - docker debug namenode
  Learn more at https://docs.docker.com/go/debug-cli/

2026-06-02 15:35:30,881 INFO sas1.sas10atransferClient: SASL encryption trust check: localHostTrusted = false, remoteHostTrusted = false

What's next:
  Try Docker Debug for seamless, persistent debugging tools in any container or image - docker debug namenode
  Learn more at https://docs.docker.com/go/debug-cli/
  Deleted /user/root/output

What's next:
  Try Docker Debug for seamless, persistent debugging tools in any container or image - docker debug namenode
  Learn more at https://docs.docker.com/go/debug-cli/

2026-06-02 15:35:35,188 WARN startLogStreamJob: file option is deprecated, please use generic option -files instead.
packageJobJar: [mapper.py, reducer.py] [] /tmp/streamJob276358823694007200.jar tmpoir=null
2026-06-02 15:35:35,734 INFO impl.metricsConfig: Loaded properties from hadoop.metrics2.properties
2026-06-02 15:35:35,773 INFO impl.metricsContextImpl: Scheduled metric snapshot period at 10 second(s).
2026-06-02 15:35:35,773 INFO impl.metricsSystemImpl: JobTracker metrics system started
2026-06-02 15:35:35,786 WARN impl.metricsSystemImpl: JobTracker metrics system already initialized!
2026-06-02 15:35:35,938 INFO mapred.FileOutputFormat: Total input files to process = 3
2026-06-02 15:35:35,963 INFO mapreduce.JobSubmitter: number of splits:1
2026-06-02 15:35:36,046 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local671748445_0001
2026-06-02 15:35:36,046 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-06-02 15:35:36,157 INFO mapred.LocalDistributedCacheManager: Localized file: job_files/mapper.py as file:/tmp/hadoop-root/mapred/local/job_local671748445_0001/diede2e5af749f5bcf81c7238f94ae/mapper.py
2026-06-02 15:35:36,169 INFO mapred.LocalDistributedCacheManager: Localized file: job_files/reducer.py as file:/tmp/hadoop-root/mapred/local/job_local671748445_0001/c1c21ab-0c43-4a4d-b060-2f1f1d07dab/reducer.py

2026-06-02 15:35:36,208 INFO mapreduce.Job: The url to track the job: http://localHost:8080/
2026-06-02 15:35:36,210 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-06-02 15:35:36,210 INFO mapreduce.Job: Running Job: local671748445_0001
2026-06-02 15:35:36,211 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapred.FileOutputCommitter
2026-06-02 15:35:36,214 INFO output.FileOutputCommitter: File OutputCommitter Algorithm version is 2
2026-06-02 15:35:36,214 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup temporary folders under output directory:false, ignore cleanup failures: false
2026-06-02 15:35:36,242 INFO mapred.LocalJobRunner: waiting for map tasks
2026-06-02 15:35:36,244 INFO mapred.LocalJobRunner: starting task: attempt_local671748445_0001_m_000000_0
2026-06-02 15:35:36,260 INFO output.FileOutputCommitter: File OutputCommitter Algorithm version is 2
2026-06-02 15:35:36,260 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup temporary folders under output directory:false, ignore cleanup failures: false
2026-06-02 15:35:36,270 INFO mapred.Task: Using ResourceCalculatorProcessTree : [ ]
2026-06-02 15:35:36,275 INFO mapred.MapTask: Processing split: hdfs://namenode:9000/user/root/input/input.txt:0:717
2026-06-02 15:35:36,286 INFO mapred.MapTask: numReduceTasks: 1
2026-06-02 15:35:36,338 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396/104857584)
2026-06-02 15:35:36,338 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-06-02 15:35:36,338 INFO mapred.MapTask: soft limit at 83886080
2026-06-02 15:35:36,338 INFO mapred.MapTask: bufstart = 0; bufused = 104857600
2026-06-02 15:35:36,338 INFO mapred.MapTask: vstarep = 26214396; length = 6353600
2026-06-02 15:35:36,341 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-06-02 15:35:36,346 INFO streaming.PipedMapred: KvpMapred exec (/job_files/_mapper.py)
2026-06-02 15:35:36,349 INFO configuration.deprecation: mapred.work.output.dir is deprecated. Instead, use mapreduce.task.output.dir
2026-06-02 15:35:36,350 INFO configuration.deprecation: map.input.start is deprecated. Instead, use mapreduce.map.input.start
2026-06-02 15:35:36,350 INFO configuration.deprecation: mapred.task.is.map is deprecated. Instead, use mapreduce.task.ismap

```

```

2026-06-02 15:35:36.341 INFO mapped.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-06-02 15:35:36.346 INFO streaming.PipedMapred: PipedMapred exec [/job_files/_mapper.py]
2026-06-02 15:35:36.349 INFO Configuration.deprecation: mapred.work.output.dir is deprecated. Instead, use mapreduce.task.output.dir
2026-06-02 15:35:36.350 INFO Configuration.deprecation: map.input.start is deprecated. Instead, use mapreduce.map.input.start
2026-06-02 15:35:36.350 INFO Configuration.deprecation: mapred.task.is.map is deprecated. Instead, use mapreduce.task.ismap
2026-06-02 15:35:36.351 INFO Configuration.deprecation: mapred.task.id is deprecated. Instead, use mapreduce.task.attempt.id
2026-06-02 15:35:36.351 INFO Configuration.deprecation: mapred.tip.id is deprecated. Instead, use mapreduce.task.id
2026-06-02 15:35:36.351 INFO Configuration.deprecation: mapred.local.dir is deprecated. Instead, use mapreduce.cluster.local.dir
2026-06-02 15:35:36.352 INFO Configuration.deprecation: map.input.file is deprecated. Instead, use mapreduce.map.input.file
2026-06-02 15:35:36.352 INFO Configuration.deprecation: mapred.skip.on is deprecated. Instead, use mapreduce.job.skiprecords
2026-06-02 15:35:36.352 INFO Configuration.deprecation: map.input.length is deprecated. Instead, use mapreduce.map.input.length
2026-06-02 15:35:36.353 INFO Configuration.deprecation: mapred.job.id is deprecated. Instead, use mapreduce.job.id
2026-06-02 15:35:36.353 INFO Configuration.deprecation: user.name is deprecated. Instead, use mapreduce.job.user.name
2026-06-02 15:35:36.354 INFO Configuration.deprecation: mapred.task.partition is deprecated. Instead, use mapreduce.task.partition
2026-06-02 15:35:36.376 INFO ssl.SslDataTransferClient: GSK encryption trust check: LocalHostTrusted = false, remoteTrusted = false
2026-06-02 15:35:36.441 INFO streaming.PipedMapred: R/W=5/1,0/0 InNA [rec/s] out:NA [rec/s]
2026-06-02 15:35:36.441 INFO streaming.PipedMapred: R/W=5/1,0/0 InNA [rec/s] out:NA [rec/s]
2026-06-02 15:35:36.442 INFO streaming.PipedMapred: Records R/W=5/1
2026-06-02 15:35:36.445 INFO streaming.PipedMapred: MRErrorThread done
2026-06-02 15:35:36.445 INFO streaming.PipedMapred: MapredFinished
2026-06-02 15:35:36.448 INFO mapped.LocalJobRunner:
2026-06-02 15:35:36.448 INFO mapped.MapTask: Starting flush of map output
2026-06-02 15:35:36.448 INFO mapped.MapTask: bufstart = 0, bufend = 417, bufvoid = 104857600
2026-06-02 15:35:36.448 INFO mapped.MapTask: kvstart = 26214396(104857584); kvend = 26214200(104856800); length = 197/653600
2026-06-02 15:35:36.454 INFO mapped.MapTask: Finished spill 0
2026-06-02 15:35:36.454 INFO mapped.Task: Task attempt_local671748445_0001_m_000000_0 is done. And is in the process of committing
2026-06-02 15:35:36.467 INFO mapped.LocalJobRunner: Records R/W=5/1
2026-06-02 15:35:36.468 INFO mapped.Task: Task attempt_local671748445_0001_m_000000_0 done.
2026-06-02 15:35:36.474 INFO mapped.Task: Final counters for attempt_local671748445_0001_m_000000_0: Counters: 23
File system counters:
FILE: Number of bytes read=2626
FILE: Number of bytes written=31973
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=217
HDFS: Number of bytes written=0
HDFS: Number of read operations=5
HDFS: Number of large read operations=0
HDFS: Number of write operations=1
HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
Map input records=50
Map output records=50
Map output bytes=417
Map output materialized bytes=523
Input split bytes=98
Combine input records=0
Spilled Records=50
Failed Shuffles=0
Merged Map outputs=0
GC time elapsed (ms)=7
Total committed heap usage (bytes)=293076992
File Input Format Counters
Bytes Read=217
2026-06-02 15:35:36.474 INFO mapped.LocalJobRunner: Finishing task: attempt_local671748445_0001_m_000000_0
2026-06-02 15:35:36.474 INFO mapped.LocalJobRunner: map task executor complete.
2026-06-02 15:35:36.477 INFO mapped.LocalJobRunner: Waiting for reduce tasks
2026-06-02 15:35:36.477 INFO mapped.LocalJobRunner: Starting task: attempt_local671748445_0001_r_000000_0
2026-06-02 15:35:36.480 INFO output.FileOutputCommitter: File OutputCommitter Algorithm version is 2
2026-06-02 15:35:36.484 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup_temporary folders under output directory:false, ignore cleanup failures: false
2026-06-02 15:35:36.484 INFO mapped.task: Using ResourceCalculatorProcressTree : 0
2026-06-02 15:35:36.484 INFO mapped.reduceTask: Using ShuffleConsumerPlugin: org.apache.hadoop.mapreduce.task.shuffle.ShuffleFetchAndMapTaskV2
2026-06-02 15:35:36.488 WARN Impl.MetricsSystemImpl: JobTracker metrics system already initialized!
2026-06-02 15:35:36.500 INFO reduce.MergerManagerImpl: MergerManager: mergeInfo:input=128307568, maxShuffleFetch=1320759392, mergeThreshold=846684800, ioSortFactor=10, memThreshold=10
2026-06-02 15:35:36.500 INFO reduce.MergerFetcher: attempt_local671748445_0001_r_000000_0 Thread state: EventFetcher for fetching map completion events
2026-06-02 15:35:36.522 INFO reduce.LocalFetcher: localFetcher#1 about to shuffle output of map attempt_local671748445_0001_m_000000_0 from 519 len: 523 to MEMORY

```

```
2026-06-02 15:35:36.484 INFO mapred.Task: Using ResourceCalculatorProcessTree: [ ]
2026-06-02 15:35:36.486 INFO mapred.ReduceTask: Using ShuffleConsumerPlugin: org.apache.hadoop.mapreduce.task.reduce.Shuffle45646c42
2026-06-02 15:35:36.488 WARN Impl.MetricsSystemImpl: Jobtracker metrics system already initialized!
2026-06-02 15:35:36.500 INFO reduce.MergeManagerImpl: MergeManager: memoryLimit=1283037568, maxSingleShuffleLimit=320759392, mergeThreshold=846804800, ioSortFactor=10, memToMemMergeOutputsThreshold=10
2026-06-02 15:35:36.502 INFO reduce.EventFetcher: attempt_local671748445_0001_r_000000_0 Thread started: EventFetcher for fetching Map completion Events
2026-06-02 15:35:36.522 INFO reduce.LocalFetcher: LocalFetcher#1 about to shuffle output of map attempt_local671748445_0001_m_000000_0 decomp: 519 len: 523 to MEMORY
2026-06-02 15:35:36.526 INFO reduce.InMemoryMapOutput: Read 519 bytes from map-output for attempt_local671748445_0001_m_000000_0
2026-06-02 15:35:36.528 INFO reduce.MergeManagerImpl: closeInMemoryFile -> map-output of size: 519, inMemoryMapOutputs.size() -> 1, commitMemory -> 0, usedMemory -> 519
2026-06-02 15:35:36.529 INFO reduce.EventFetcher: EventFetcher is interrupted.. Returning
2026-06-02 15:35:36.530 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-06-02 15:35:36.530 INFO reduce.MergeManagerImpl: finalMerge called with 1 in-memory map-outputs and 0 on-disk map-outputs
2026-06-02 15:35:36.535 INFO mapred.Merger: Merging 1 sorted segments
2026-06-02 15:35:36.536 INFO mapred.Merger: Down to the last merge-pass, with 1 segments left of total size: 515 bytes
2026-06-02 15:35:36.537 INFO reduce.MergeManagerImpl: Merged 1 segments, 519 bytes to disk to satisfy reduce memory limit
2026-06-02 15:35:36.538 INFO reduce.MergeManagerImpl: Merging 1 files, 523 bytes from disk
2026-06-02 15:35:36.539 INFO reduce.MergeManagerImpl: Merging 0 segments, 0 bytes from memory into reduce
2026-06-02 15:35:36.539 INFO mapred.Merger: Merging 1 sorted segments
2026-06-02 15:35:36.540 INFO mapred.Merger: Down to the last merge-pass, with 1 segments left of total size: 515 bytes
2026-06-02 15:35:36.541 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-06-02 15:35:36.544 INFO streaming.PipeMapRed: PipeMapRed: exec [/job_files/./reducer.py]
2026-06-02 15:35:36.545 INFO Configuration.deprecation: mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-06-02 15:35:36.546 INFO Configuration.deprecation: mapred.map.tasks is deprecated. Instead, use mapreduce.job.maps
2026-06-02 15:35:36.572 INFO streaming.PipeMapRed: R/W/S=1/0/0 in:NA [rec/s] out:NA [rec/s]
2026-06-02 15:35:36.573 INFO streaming.PipeMapRed: R/W/S=10/0/0 in:NA [rec/s] out:NA [rec/s]
2026-06-02 15:35:36.576 INFO streaming.PipeMapRed: Records R/W=50/1
2026-06-02 15:35:36.578 INFO streaming.PipeMapRed: MERGERead done
2026-06-02 15:35:36.578 INFO streaming.PipeMapRed: mapred-finished
2026-06-02 15:35:36.593 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localhostTrusted = false, remoteHostTrusted = false
2026-06-02 15:35:36.611 INFO mapred.Task: Task attempt_local671748445_0001_r_000000_0 is done. And is in the process of committing
2026-06-02 15:35:36.613 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-06-02 15:35:36.613 INFO mapred.Task: Task attempt_local671748445_0001_r_000000_0 is allowed to commit now
2026-06-02 15:35:36.625 INFO output.FileOutputCommitter: Saved output of task 'attempt_local671748445_0001_r_000000_0' to hdfs://namenode:9000/user/root/output
2026-06-02 15:35:36.626 INFO mapred.LocalJobRunner: records R/W=50/1 > reduce
2026-06-02 15:35:36.626 INFO mapred.Task: Task 'attempt_local671748445_0001_r_000000_0' done.
2026-06-02 15:35:36.627 INFO mapred.Task: Final Counters for attempt_local671748445_0001_r_000000_0: Counters: 30
  File system Counters
    FILE: Number of bytes read=3704
    FILE: Number of bytes written=532496
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=717
    HDFS: Number of bytes written=57
    HDFS: Number of read operations=10
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=3
    HDFS: Number of bytes read erasure-coded=0
  Map-Reduce Framework
    Combine input records=0
    Combine output records=0
    Reduce input groups=5
    Reduce shuffle bytes=523
    Reduce input records=50
    Reduce output records=5
    Spilled records=10
    Shuffled Maps =1
    Failed Shuffles=0
    Merged Map outputs=1
    GC time elapsed (ms)=0
    Total committed heap usage (bytes)=293076992
  Shuffle Errors
    BAD_ID=0
    CONNECTION=0
    IO_ERROR=0
    WRONG_LENGTH=0
    WRONG_MAP=0
    WRONG_REDUCE=0
  File Output Format Counters
    Bytes Written=57
```

```

        Shuffled Maps =1
        Failed Shuffles=0
        Merged Map outputs=1
        GC time elapsed (ms)=0
        Total committed heap usage (bytes)=293076992
    Shuffle Errors
        BAD_ID=0
        CONNECTION=0
        IO_ERROR=0
        WRONG_LENGTH=0
        WRONG_MAP=0
        WRONG_REDUCE=0
    File Output Format Counters
        Bytes Written=57
2026-06-02 15:35:36,627 INFO mapred.LocalJobRunner: Finishing task: attempt_local671748445_0001_r_000000_0
2026-06-02 15:35:36,627 INFO mapred.LocalJobRunner: reduce task executor complete.
2026-06-02 15:35:37,215 INFO mapreduce.Job: Job job_local671748445_0001 running in uber mode : false
2026-06-02 15:35:37,215 INFO mapreduce.Job: map 100% reduce 100%
2026-06-02 15:35:37,216 INFO mapreduce.Job: Job job_local671748445_0001 completed successfully
2026-06-02 15:35:37,222 INFO mapreduce.Job: Counters: 36
    File System Counters
        FILE: Number of bytes read=6330
        FILE: Number of bytes written=1064469
        FILE: Number of read operations=0
        FILE: Number of large read operations=0
        FILE: Number of write operations=0
        HDFS: Number of bytes read=1434
        HDFS: Number of bytes written=57
        HDFS: Number of read operations=15
        HDFS: Number of large read operations=0
        HDFS: Number of write operations=4
        HDFS: Number of bytes read erasure-coded=0
    Map-Reduce Framework
        Map input records=50
        Map output records=50
        Map output bytes=417
        Map output materialized bytes=523
        Input split bytes=98
        Combine input records=0
        Combine output records=0
        Reduce input groups=5
        Reduce shuffle bytes=523
        Reduce input records=50
        Reduce output records=5
        Spilled Records=100
        Shuffled Maps =1
        Failed Shuffles=0
        Merged Map outputs=1
        GC time elapsed (ms)=7
        Total committed heap usage (bytes)=586153984
    Shuffle Errors
        BAD_ID=0
        CONNECTION=0
        IO_ERROR=0
        WRONG_LENGTH=0
        WRONG_MAP=0
        WRONG_REDUCE=0
    File Input Format Counters
        Bytes Read=717
    File Output Format Counters
        Bytes Written=57
2026-06-02 15:35:37,222 INFO streaming.StreamJob: Output directory: /user/root/output

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image - docker debug namenode
    Learn more at https://docs.docker.com/go/debug-cli/

boomr@DESKTOP-IU0AVN0 MINGW64 ~/OneDrive/Documents/GitHub/DADS6002/code (main)
$

```

3.ผลลัพธ์ MapReduce

- MSYS_NO_PATHCONV=1 docker exec -it namenode hdfs dfs -cat /user/root/output/part-00000

MapReduce Output

```
boomn@DESKTOP-IU0AVN0 MINGW64 ~/OneDrive/Documents/GitHub/DADS6002/code (main)
$ MSYS_NO_PATHCONV=1 docker exec -it namenode hdfs dfs -cat /user/root/output/part-00000
2026-06-02 15:24:19,931 INFO sas1.SaslDataTransferClient: SASL encryption trust check: localHostTrusted = false, remoteHostTrusted = false
1      163214.29
2      103600.00
3      59300.00
4      30571.43
5      20888.89

what's next:
  Try Docker Debug for seamless, persistent debugging tools in any container or image - docker debug namenode
  Learn more at https://docs.docker.com/go/debug-cli/

boomn@DESKTOP-IU0AVN0 MINGW64 ~/OneDrive/Documents/GitHub/DADS6002/code (main)
$ |
```